Multispectral Texture Synthesis Using RGB Convolutional Neural Networks

Sélim Ollivier, Yann Gousseau, and Sidonie Lefebvre

***Abstract*— State-of-the-art red–green–blue (RGB) texture synthesis algorithms rely on *style distances* that are computed through statistics of deep features. These deep features are extracted by classification neural networks that have been trained on large datasets of RGB images. Extending such synthesis methods to multispectral images is not straightforward, since the pretrained networks are designed for and have been trained on RGB images. In this work, we propose two solutions to extend these methods to multispectral imaging (MSI). Neither of them requires additional training of the neural network from which the second-order neural statistics are extracted. The first one involves optimizing over batches of random triplets of spectral bands during training. The second one projects multispectral pixels onto a 3-D space. We further explore the benefit of a color transfer operation upstream of the projection to avoid the potentially abnormal color distributions induced by the projection. Our experiments compare the performances of the various methods through different metrics. We demonstrate that they can be used to perform exemplar-based texture synthesis, achieve good visual quality, and come close to state-of-the-art methods on RGB bands. Code is available at https://github.com/selim2483/multispectral-textureCNN.**

***Index Terms*— Multispectral imaging (MSI), texture synthesis.**

I. Introduction

In RECENT years, a great deal of effort has been devoted to the development and design of multispectral imagers, capable of capturing images of a scene in multiple (usually between 2 and 20) spectral bands in the infrared or visible range. These devices serve various purposes, such as detecting aircraft and autonomous aerial vehicles and analyzing satellite data for Earth observation, including vegetation, atmosphere, ocean, and land assessments. Moreover, numerous deep learning methods have been developed in response to growing interest in multispectral imaging (MSI) and its applications.

Yet, the training of efficient classifiers and detection systems requires access to a large amount of data [1]. Similarly, the augmentation of existing databases is highly beneficial for the performance assessment of image processing algorithms and optical sensors [2] and for enhancing generalization capacity [3]. In both cases, it is necessary to have reference scenes and to be able to model the variability of objects of interest and background environments. Given the significant costs associated with data acquisition and annotation, the ability to rapidly synthesize images that accurately replicate the radiometric properties and textures of real-world backgrounds is highly advantageous. Indeed, large amounts of Earth observation data are freely accessible from some multispectral satellite missions, such as Sentinel-2, but only for certain spectral band combinations and spatial resolutions. It is even more difficult to have access to large quantities of data for ground sensors, which are often used for site monitoring applications, such as drone detection [4]. Synthetic multispectral data thus enables to expansion of small datasets. When assessing new multispectral sensor designs prior to expensive hardware prototyping, it can be valuable to evaluate the contribution of different spectral bands and spatial resolutions for specific applications. Additionally, synthetic data can help balance existing datasets, providing more examples of minority classes to improve the performance of algorithms on these classes, which can be of great importance. Generative models also allow for the synthesis of variations of existing data, in a controlled way [2], [5]. A concrete application of this can be found in cloud detection and removal applications. In that context, one needs to have access to pairs of images of the same scene with and without clouds to perform cloud removal, as it is done by [6]. Being able to control the synthesis can also enable one to focus on different types of natural backgrounds for developing and testing image processing and analysis algorithms overcoming real-world data limitations. For example, one could generate clouds that maintain the same overall coverage as a prototype image while exhibiting different spatial textures, providing a more flexible framework for algorithm evaluation.

Texture modeling and analysis were initially limited to monoband images for which tools, such as local statistics extraction, co-occurrence matrices, wavelet transforms, or fractal-based techniques, have been developed to account for spatial behavior and patterns. Extensions of these techniques to color and multispectral/hyperspectral images followed afterward. In this context, the color or spectral statistics need to be taken into account. For this purpose, dimensionality reduction [7] or 3-D power spectral density and histogram matching [8] have been introduced for texture characterization or synthesis.

Since the foundational work of Gatys et al [9], pretrained convolutional neural networks (CNNs) have been at the core of the state-of-the-art in the field of exemplar-based texture synthesis for natural images. Most subsequent works used statistics extracted from the feature maps of these networks

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Sélim Ollivier is with LTCI, Télécom Paris, Institut Polytechnique de Paris, 91120 Palaiseau, France, and also with DOTA/AILab, ONERA, Université Paris-Saclay, 91120 Palaiseau, France (e-mail: selim.ollivier@onera.fr).

Yann Gousseau is with LTCI, Télécom Paris, Institut Polytechnique de Paris, 91120 Palaiseau, France.

Sidonie Lefebvre is with DOTA/AILab, ONERA, Université Paris-Saclay, 91120 Palaiseau, France.

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to model texture through *style distances*. This approach is particularly effective because it capitalizes on the network's powerful description and recognition abilities, which stem from its architecture and its training on large datasets. Recently, generative networks for the synthesis of satellite data and multispectral/hyperspectral images have emerged. These models are often based on architectures that have been proven to be effective for natural images and manage to generate realistic multispectral images by training new neural networks on multispectral/hyperspectral images in an adversarial [10], [11] or diffusion-based [12] framework. However, to the best of our knowledge, no method has been proposed to perform multispectral/hyperspectral texture synthesis that leverages style distances relying on statistics extracted from pretrained networks as those used for natural images. Indeed, this is not straightforward, since the widely used VGG19 network [13], trained on ImageNet, is designed for three-channel images since natural images are coded according to the red–green–blue (RGB) model. Although interesting works have tried to bypass pretrained networks such as VGG-19 to define style distances, results remain of lower quality [14]. Additionally, more advanced architectures such as ResNets [15], which perform better on classification tasks, also fail to outperform VGG in texture synthesis and style transfer because of the inability of the residual layer to capture high-level details through Gram matrices [16]. Therefore, VGG networks are a key ingredient in defining style distances and design methods for texture-related tasks for RGB images. Besides, training or finetuning a new network with a VGG-like architecture for MSI would require access to large multispectral/hyperspectral databases.

To avoid this, the present work introduces two different strategies to build multispectral style distances based on pretrained CNN. These distances are then successfully used to perform multispectral texture synthesis. Our approach can be applied to a wide range of natural textures. Our primary focus is on cloud backgrounds that are useful for various applications: cloud cover forecasting to inform acquisition decisions, and thus optimize the planning of agile satellite constellations for Earth observation, but also to predict the availability of optical communication links between ground stations and satellites. Additional use cases involve the detection of small objects, such as drones, against a clouded sky background or cloud removal in optical satellite images [6]. We also extend our experiments to a more general range of textures with the HyTexiLa database [17].

Our contributions can be summarized as the following.

1) We introduce two style distances: a stochastic style distance and a projected style distance that can be computed from a pretrained RGB CNN. This in turn allows to synthesis of multispectral images using an optimization process similar to the one of [9]. Our approach is not specific to texture modeling and could also be used to extend other distances based on pretrained RGB networks to MSI without having to train new networks for that purpose.
2) To correct the color domain shift possibly induced by the projection when using the projected style distance, we propose to rely on color transfer. This operation is meant to align the projected image with a more typical color distribution seen during training by the pretrained CNN involved in the style distance. We then empirically investigate its usefulness.

Section II reviews related work on multispectral texture modeling, and Section III presents our proposed methodology in detail. Texture synthesis experiments on 130 multispectral cloud field images from the Sentinel-2 mission and on 112 multispectral texture images from the HyTexiLa database are conducted in Section IV. We conclude this article in Section V.

II. Related Work

In this section, we present some methods related to our work. We first discuss the various techniques used in the literature to model texture in multispectral and hyperspectral images. Then, we introduce recent advances in deep generative models for multispectral and hyperspectral imaging and finally give an overview of the literature on neural texture synthesis.

A. Multispectral and Hyperspectral Texture Modeling

Texture synthesis and modeling first emerged in applications linked to monochromatic images and extensions to color images were proposed later on. For color texture synthesis, considering each color channel independently is often not satisfying due to interchannel correlations. Early methods for color texture synthesis apply a principal component analysis (PCA) before working channel-by-channel [18], possibly modeling interchannel correlations as in [19]. For hyperspectral images, Mercier and Lennon [7] use an independent component analysis before applying wavelet decomposition for texture characterization. Sarkar and Healey [8] propose to constrain the multidimensional histograms of images together with a 3-D Fourier transform accounting for intraband spatial dependency to perform texture synthesis.

From a modeling perspective, textures were originally often analyzed through dependencies between nearby pixels, which led to the use of Markov models. Adding the assumption of a Gaussian response for each vertex of the graph representing the image to be analyzed led to a model well-known as a random Gauss–Markov field. These models were first leveraged for anomaly detection in multispectral and hyperspectral images by Schweizer and Moura [20], [21]: the interpixel Markov component allows textures to be taken into account while the underlying Gaussian assumption allows a simple modeling of the spectral response. Some Gauss–Markov models with a different way of accounting for second-order spectral information were proposed by Rellier et al. [22], [23], for hyperspectral textures classification. All these models are based on sets of simplifying assumptions taking into account the properties of the hyperspectral images analyzed. These assumptions relate in particular to the constancy across the spectrum of the Markov parameters governing the textures and the spectral covariance modeling of the textures across the spectrum. Spatial parameters are typically assumed to be constant throughout the spectrum, which means that the

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multispectral texture is considered to be identical at each wavelength, up to a factor of variance. The Schweizer and Moura formulation implies that each scalar pixel interacts with its neighbors in all three dimensions (spatial and spectral) and the inclusion of an interband Markov link in the model implicitly imposes identical interband behavior between all pairs of neighboring bands. Rellier et al proposed to restrict the pixel neighborhood to spatial neighbors. The aim is then to model the spatial interactions between hyperpixels, and thus the expression of the spatial part of the precision matrix loses its interband component. Although more flexible variations have been proposed, these underlying simplifying assumptions limit the expressiveness of the textures generated, and texture synthesis methods based on generative networks offer a promising alternative.

To assess the quality of the synthesis, conventional texture metrics, such as local binary pattern and co-occurrence matrices, have been exploited, with within and between channel interactions, in a recent comparison of color imaging and hyperspectral imaging for texture classification [24]. Recently, Chu and co-authors focused on accounting simultaneously for spectral and spatial distribution. They proposed a 4-D texture metric, GHOST [25], encompassing joint metrological assessment of spectral and spatial properties. They also conceived an alternative feature, the relative spectral difference occurrence matrix (RSDOM) [26], [27], which considers the joint probability of spectral distribution and pixel spatial arrangement. It consists of pixel differences relative to a spectral reference and spatial neighbors, emphasizing its contribution to the hyperspectral image texture classification scheme on the HyTexiLa database [17].

B. Deep Generative Models for Multispectral and Hyperspectral Images

The most recent generative neural network methods, such as generative adversarial networks (GANs) and diffusion models, have only very recently started to be applied to multispectral satellite data. The recent work of [11] compares, both with a spectral signature analysis and a modified Fréchet inception distance, the performances of StyleGAN3 and proGAN architectures to reproduce Sentinel-2 images. Khanna et al. [28] propose DiffusionSat, a generative foundation model for remote sensing data based on the latent-diffusion model architecture of StableDiffusion [29], dedicated to inverse problems, such as multispectral input super-resolution, temporal prediction, and in-painting. To alleviate the computational complexity due to the high spectral dimensionality, UnmixDiff [2] has recently been proposed, and consists of coupling an unmixing autoencoder and a diffusion model. The synthesis is thus performed in the abundance space, which enables to obtaining realistic objects' distribution in abundance maps. However, this method focuses on the spectral profile consistency rather than on the texture representativity of the different natural background components, such as forest, sky, grassland... Another publication [12] emphasizes the use of a diffusion model for the synthesis of hyperspectral images, but with a conditional RGB image as input. They introduce a conditional vector quantized generative adversarial network (VQGAN) to map hyperspectral data into a low-dimensional latent space, in which the diffusion model is easier to train. They pay particular attention to the diversity of the generated hyperspectral images, by proposing two new metrics focusing on spectral content diversity. Additionally, foundation models—trained on vast unlabeled datasets and finetuned for specific tasks using limited annotated data—have seen increased use in satellite imagery over the past two years. FG-MAE [30] and SatMAE [31] have been developed for multispectral Sentinel-2 images, and DOFA [32] has been very recently proposed to adaptively integrate various data modalities into a single framework, leveraging the concept of neural plasticity in brain science. These foundation models will undoubtedly contribute to strengthening multispectral image analysis capabilities. These models have only been applied on satellite data, and mostly on data taken over cities, not on textured natural backgrounds.

C. Neural Texture Synthesis

The work from Gatys et al. [9] was a major breakthrough for exemplar-based texture synthesis. The main idea of this work, which we will detail in Section III-A, is to constrain some statistics on deep image features extracted by classical classification networks that have been trained on large image datasets (such as the VGG19 network trained on ImageNet). More precisely, second-order statistics between deep features are computed on an exemplar image through Gram matrices, and images are then synthesized by optimizing the pixel responses to match these matrices. This results in very realistic synthesized texture images, although at the price of a relatively heavy optimization procedure for each synthesis. Many works have followed, all exploiting the central idea of measuring the distance between texture images using the squared difference of their Gram matrices—distances that we will refer to as *style distances* in this article. A particular interest was devoted to developing feed-forward architecture to speed-up the synthesis [33], use the style distances in the GAN framework [34], or develop autoencoder texture synthesizers [35]. To the best of our knowledge, none of these state-of-the-art texture synthesis approaches have been adapted to the multispectral or hyperspectral framework, which is the purpose of the present article.

III. Methods

A. Neural Statistics for Texture Synthesis

As recalled in Section II, the method from Gatys et al. [9] relies on the use of statistics of deep features. These features are classically obtained from the VGG19 network. It consists of a succession of convolutional layers. Each layer l is made of N_l convolutional filters and provides a feature map $F^l \in \mathbb{R}^{N_l \times M_l}$ (where M_l is the number of pixels of the feature map). The Gram matrices were chosen by the authors to form a summary statistic that is insensitive to spatial information

$$\forall l \in \{1, \ldots, L\}, \quad G^l = F^l \left(F^l\right)^T. \tag{1}$$

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The set $G = \{G^1, \ldots, G^L\}$ of Gram matrices computed from the network in response to an image constitutes a stationary multiscale description of the texture. It can further be used to define a style distance between two images I and $\hat{I}$ with Gram matrices G and $\hat{G}$

$$\mathcal{L}_{\text{style}}(I, \hat{I}) = \sum_l w_l \left(G^l - \hat{G}^l\right)^2 \tag{2}$$

where the w_l are chosen weighting factors.

This style distance was used in [9] to achieve a clear breakthrough for texture synthesis. It was also later used to train other neural networks as a loss function [33], [35], [36]. In this work, we follow the exemplar synthesis process presented by [9]: given an exemplar image I, a synthetic image $\hat{I}$ is initialized with Gaussian white noise and then gradient descent is used to minimize the style distance between the two images I and $\hat{I}$. The result of this optimization is expected to match the Gram matrices of the exemplar. In the following, we chose to initialize the synthetic image following a Gaussian noise with the mean and covariance of the original image as it showed better performances.

This style distance makes use of the Gram matrices to describe the feature maps distributions, but other statistics such as covariance matrices (i.e., Gram matrices of the centered feature maps) have also shown good performances for texture synthesis. In the remaining of this article, we will use covariance matrices instead of Gram matrices.

The main obstacle to the extension of CNN-based texture synthesis processes to MSI is the design and the training of a multispectral CNN from which to extract feature maps statistics. To address this, we introduce a stochastic style distance in Section III-B and a style distance that makes use of projection in Section III-C. In Section III-D, we add a color operation to the projection to align the color distributions with those of the images the network has been trained on. These distances enable one to use the VGG19 network, pretrained on ImageNet, which has become standard for three-channel images.

B. Stochastic Style Distance

The difficulty in extending the approach recalled in Section III-A to multispectral images is to constrain the statistics of an image of dimension N with an optimization procedure that relies on an RGB CNN. To do so, we decided to constrain the statistics of all possible three-channel images obtained from the channels of the multispectral image.

We consider an exemplar multispectral image I with N spectral bands and $\hat{I}$ its synthetic counterpart. We propose to randomly draw a triplet J of integers in $\{1, \ldots, N\}$ at each iteration of the optimization. Then, to evaluate the style distance $\mathcal{L}_{\text{style}}(I_J, \hat{I}_J)$ between the images I_J and $\hat{I}_J$, which are composed of the spectral bands indexed by the integers of J. Finally, the gradient of this quantity is used to take an optimization step on $\hat{I}$.

During the optimization, one can average the style distance over batches of uniformly drawn triplets in a way that is reminiscent of stochastic gradient descent algorithms. It ensures to take into account more spectral bands and correlations at each iteration. Moreover, it regularizes the objective function.

The global minimization objective of our algorithm is the expectation of the style distance between three-channel images corresponding to the spectral bands indexed by a triplet J that follows the uniform law. This amounts to averaging the style distance over all possible triplets

$$\mathcal{L}^{\text{MS}}_{\text{style}}(I, \hat{I}) = \mathbb{E}_{J \sim \mathcal{U}(\mathcal{J}_N)}\left[\mathcal{L}_{\text{style}}(I_J, \hat{I}_J)\right] \tag{3}$$

$$= \frac{1}{N_J} \sum_{J \in \mathcal{J}_N} \mathcal{L}_{\text{style}}(I_J, \hat{I}_J) \tag{4}$$

where:

1) $\mathcal{J}_N$ is the set of triplets of integers in $\{1, \ldots, N\}$ of cardinal N_J
2) $\mathcal{U}(\mathcal{J}_N)$ is the uniform distribution on $\mathcal{J}_N$
3) $\mathbb{E}_{J \sim \mathcal{U}(\mathcal{J}_N)}$ is the mathematical expectation under the condition that J follows a uniform distribution
4) If J is a triplet, I_J is the three-channel image made of the 3 spectral bands of I indexed by J
5) $\mathcal{L}_{\text{style}}$ is the classical style loss for RGB images using a VGG19 pretrained network described by [9], recalled in (2).

The proposed synthesis algorithm is detailed in Algorithm 1 for n iterations of the gradient descent algorithm and B triplets drawn at every iteration. The mean and covariance of I are denoted as μ_I and Σ_I.

Algorithm 1 Texture Synthesis With Stochastic Style Loss

```
Î ~ N(μ_I, Σ_I)                          ▷ Gaussian noise initialization
for i ∈ {1, ..., n} do
    L ← 0
    for j ∈ {1, ..., B} do
        J ~ U(J_N)                       ▷ Uniformly drawn triplet
        L ← L + L_style(I_J, Î_J)
    end for
    Take a gradient step on Î with gradient ∇L
end for
return Î
```

Our method ensures that browse every spectral band and consider the correlations between each of them in the course of the optimization. However, a drawback of this method is that it may need a relatively large number of triplets to reach a good overall control over multispectral statistics, resulting in a high-computational cost. In Section III-C, we introduce a more simple and faster alternative approach.

C. Projected Style Distance

The second style distance that we propose aims to take advantage of the redundancy of the spectral information. Indeed, the spectral information that is contained in N dimensions often can be reduced to a lower number of dimensions. Therefore, we consider a projector $P: \mathbb{R}^N \to \mathbb{R}^3$ that maps multispectral images with N spectral bands into RGB-like images with three channels. The proposed style distance consists then in the classical style distance computed on projected

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images

$$\mathcal{L}^{P}_{\text{style}}\big(I, \hat{I}\big) = \mathcal{L}_{\text{style}}\big(P(I), P(\hat{I})\big). \tag{5}$$

The use of the projected style distance for exemplar-based texture synthesis is detailed in Algorithm 2.

Algorithm 2 Texture Synthesis With Projected Style Loss

$\hat{I} \sim \mathcal{N}(\mu_I, \Sigma_I)$ ▷ Gaussian noise initialization
for $i \in \{1, \ldots, n\}$ **do**
  $I_P \leftarrow P(I)$
  $\hat{I}_P \leftarrow P(\hat{I})$
  $\mathcal{L} \leftarrow \mathcal{L}_{style}\big(I_P, \hat{I}_P\big)$
  Take a gradient step on $\hat{I}$ with gradient $\nabla\mathcal{L}$
end for
return $\hat{I}$

D. Color Transfer for Neural Statistics Extraction

The projected distance introduced in Section III-C can be defined for any projection from $\mathbb{R}^N$ to $\mathbb{R}^3$. Ideally, the projection should be able to compress multispectral data while limiting the loss of information. To that end, PCA is a good candidate as selecting the first components minimizes the quadratic reconstruction error. Nevertheless, we observed that this transform can lead to unusual color distributions. The resulting domain shift can affect the quality of the description provided by the CNN as it may not have been trained on such distributions. Examples of the color that are induced by a PCA projection for cloud field images taken from the Sentinel-2 mission are available in Fig. 1.

To circumvent this, we propose to perform a color transfer operation after the projection and before the use of the VGG-19 network. The target distribution of this transfer operation is simply the color distribution of a natural image, that therefore acts as a color palette. The idea here is to mitigate the domain shift by sending back the considered three-channel image in regions of the color space that are likely to be closer to the regions that have been explored during the training of the network.

In fact, our objective is not to execute an exact color match with the palette image. Rather, our aim is to impose a color distribution that was observed during the training. To this end, we only use a coarse color transfer operation that relies on first- and second-order statistics to describe the color distributions of the images.

Hence, we consider an affine transformation that maps the projected image I_P with color statistics (μ_P, Σ_P) to an image matching the color statistics (μ_C, Σ_C) of a palette image I_C. We denote $T_{P\to C}$ such a transformation and, for an image I, we have

$$T_{P\to C}(I) = \mu_C + L_C L_P^{-1}(I - \mu_P) \tag{6}$$

where the L_P and L_C matrices are obtained using Cholesky decomposition of covariances matrices

$$\Sigma_P = L_P L_P^T \tag{7}$$
$$\Sigma_C = L_C L_C^T. \tag{8}$$

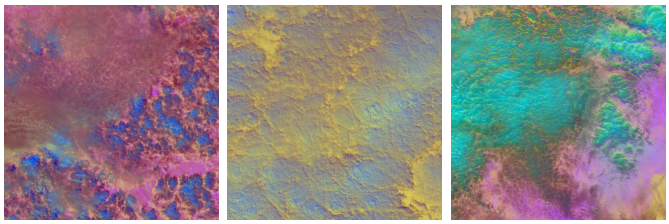


Fig. 1. Examples of unusual color distributions induced by the PCA projection of Sentinel-2 cloud field images.

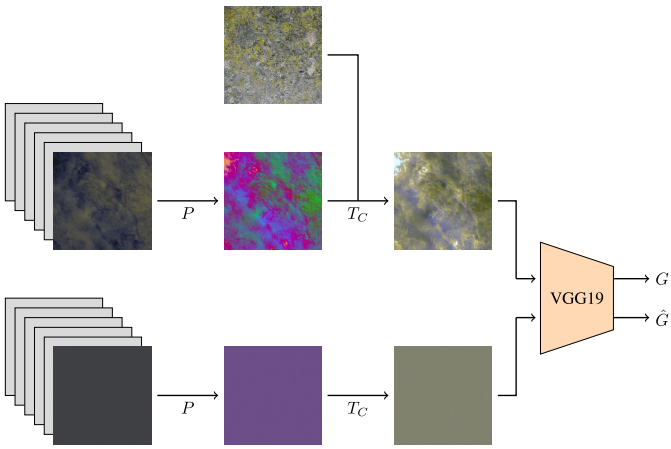


Fig. 2. Scheme of the proposed use of color transfer for multispectral texture synthesis. A projection P is used to obtain three-channel images from the exemplar and synthetic multispectral images. Then, a color transfer operator T_C is applied to both images to match their color statistics to the one of a reference palette image. Finally, statistics of deep features are extracted from the so-obtained images to perform texture synthesis.

As it is possible to define the converse transformation $T_{C\to P}$, such a transformation happens to be reversible so that it conserves the spatial information of the images and only act on the color of the pixels to match the first- and second-order statistics.

The corresponding style distance is therefore (see Fig. 2 for illustration)

$$\mathcal{L}^{P+C}_{\text{style}}\big(I, \hat{I}\big) = \mathcal{L}_{\text{style}}\big(T_{P\to C}(P(I)), T_{P\to C}(P(\hat{I}))\big) \tag{9}$$
$$= \mathcal{L}_{\text{style}}\big(P_C(I), P_C(\hat{I})\big) \tag{10}$$

with $T_{P\to C}$ being the color transformation introduced earlier, that maps the color statistics of the projection of the exemplar texture $P(I) = I_P$ into the color statistics of the palette color image I_C, and $P_C = T_C \circ P$ being the new projection operator.

E. Multispectral Texture Synthesis With Deep Features Statistics

To perform multispectral texture synthesis, we follow the same optimization procedure as in [9] and we rely on our multispectral style distances. We initialize a synthetic multispectral image of N-channels with a Gaussian noise of mean and covariance of the exemplar image as it showed better results than a white noise initialization. Then, we use L-BFGS [37] to update the synthetic multispectral image to minimize a style distance. At each iteration of the algorithm, the gradients of one of the aforementioned style distances are used to update the pixels of the N-channels synthetic image.

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TABLE I
INDEXES AND CENTRAL WAVELENGTH OF THE SELECTED BANDS FROM THE HYTEXILA DATASET

Index	6	19	33	46	56	66	96	123	150	177
Central wavelength (nm)	421.33	462.82	507.50	548.99	580.91	612.83	708.58	794.75	880.93	967.10

As mentioned before, we use covariance matrices instead of the Gram matrices for defining style losses.

IV. EXPERIMENTS

In this section, we give a brief description of the datasets we consider, the experimental settings, and the metrics used to compare the texture syntheses. Comprehensive comparisons of syntheses using the different style distances that we formerly introduced as objective functions are then presented and discussed. Finally, additional material that explores the influence of certain hyperparameters is included.

A. Datasets

1) Sentinel2: We consider a dataset composed of 130 512 × 512 Sentinel-2 images of cloud fields. The Sentinel-2 mission is an Earth observation mission from the Copernicus Program that acquires optical imagery at a spatial resolution ranging from 10 to 60 m over land and coastal waters. It captures images in 13 spectral channels in the visible/near-infrared (VNIR) and short-wave infrared spectral range (SWIR), including 9 proper observation bands and 3 atmospheric correction bands (near 442, 945, and 1375 nm), for which the atmospheric absorption is high. We operate at the L2A data level and we remove 2 of the 13 bands: B8A, which overlaps with B8 in the NIR, and the last band dedicated to atmospheric correction at 1375 nm (B10), due to its particularly high absorption making it unexploitable. The synthesis experiment was thus conducted on a dataset constituted of 11 spectral bands and on a dataset constituted of 9 spectral bands, excluding all the bands dedicated to atmospheric correction. The first dataset of 130 images of Sentinel-2 images with 11 bands is denoted as S2-11B and the second with the same 130 images and with 9 bands is denoted as S2-9B.

2) HyTexiLa: We also consider the HyTexiLa dataset [17]. It consists of 112 reflectance images of hyperspectral textures. Each image is made of 1024 × 1024 pixels and 186 spectral bands ranging from 405.37 to 995.83 nm at 3.19 nm intervals. We downsample the images to a 512 × 512 resolution by extracting a central crop. We also restrict our study to the 10 bands given in Table I.

B. Experimental Settings

We used the different style distances that we introduced in this article as objective functions to perform texture synthesis on the multispectral images.

The experimental choices and hyperparameters for each objective are as follows.

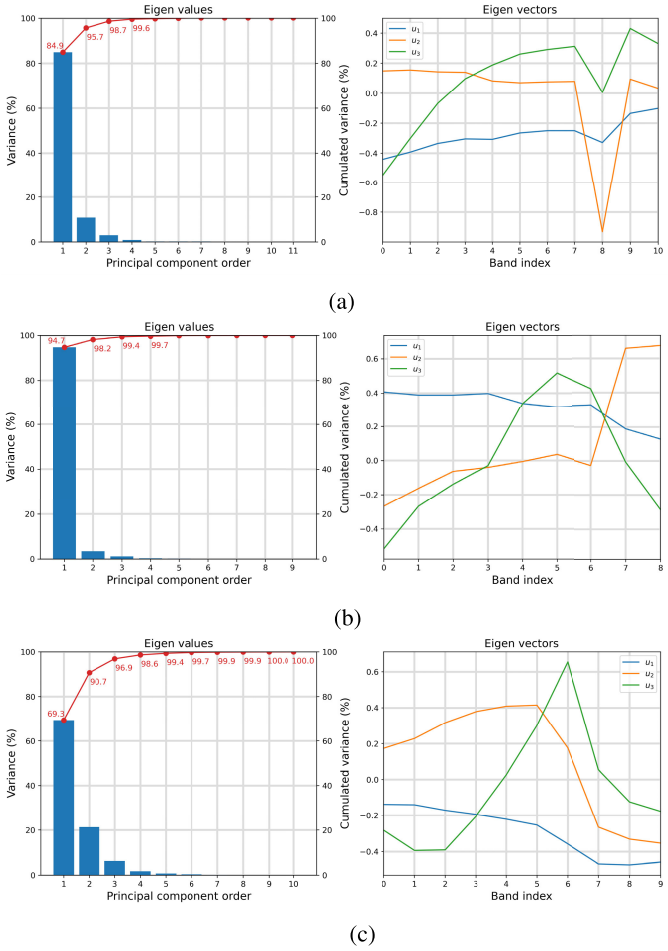


Fig. 3. Left: Eigenvalues of the PCA over the different datasets. The first component encompasses 69%–95% of the variance, the 3 first 97%–99%. Right: 3 first eigenvectors of the PCA over the different datasets (denoted as u_i). (a) Sentinel2-11B. (b) Sentinel2-9B. (c) HyTexiLa.

1) *Stochastic:* we use a batch size of ten triplets at each iteration. We show further experiments that vary the batch size in Section IV-E1.
2) *PCA:* we use PCA to encode the spectral information contained in N bands into 3-D. The first three components are sufficient to explain 99% of the variance for the S2-11B and S2-9B datasets and 97% for the HyTexiLa dataset (see Fig. 3 for eigenvalues and eigenvectors).
3) *PCA + Color:* we use the *Pebbles* image as a color palette for the color transfer operation performed after the PCA projection. We present additional experiments in which other images are used as a color palette in Section IV-E2.

For style distances, we use a VGG19 network initialized with the pretrained weights from [9]. Following the settings of this article, we extract the feature maps of the first convolutionnal filters at each spatial resolution (i.e., Conv layers

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with indices 1, 3, 5, 8, 11) and use quadratic decaying loss weights $w_l = (1/N_l^2)$ where N_l is the number of feature maps of the layer l. As mentioned in Section III-A, we use the covariance matrices as deep features statistics instead of the Gram matrices and we initialize the synthetic image with Gaussian noise with the mean and covariance of the exemplar image. We use L-BFGS for 500 iterations to perform optimization.

C. Metrics

Examplar-based texture synthesis and reconstruction differ from traditional pixel-by-pixel reconstruction, for which the syntheses can be evaluated using pixel-wise metrics, such as the root mean square error (RMSE), spectral angle mapper (SAM), or peak signal-to-noise ratio (PSNR). Here, we want to generate a new texture image that conserves the statistics, the properties, and the general visual aspect of the original image while allowing another spatial distribution. Therefore, traditional pixel-wise metrics such as the PSNR are not adapted. In the following, we introduce different metrics used to evaluate the performance of our methods.

1) Style: We measure different style distances between original and synthetic images.

1) The stochastic style distance denoted as $\mathcal{L}^{\text{MS}}_{\text{style}}$. It is computed by averaging over every possible triplet. This is precisely the metric corresponding to the stochastic style distance.
2) The projected style distance using a PCA, denoted as $\mathcal{L}^{\text{PCA}}_{\text{style}}$, corresponds to the second style distance introduced in this article.
3) The style distance for the bands that are the closest to RGB wavelengths (4, 3, and 2 for Sentinel2 and 5, 3, and 1 for Hytexila). It is denoted as $\mathcal{L}^{\text{RGB}}_{\text{style}}$.

2) Multispectral Distribution: In the context of RGB images, color distribution is a central feature in texture description. For multispectral images, we similarly want to synthesize images having a distribution of pixel values as close as possible to the exemplar image. An efficient way to compare multidimensional distribution is the Wasserstein distance. Because it is very costly to compute such distances in dimensions greater than two, we either rely on the sliced Wasserstein distance (SWD) [38] or on band-wise Wasserstein distances.

1) SWD $\mathcal{L}_{\text{hist}}$

$$\begin{aligned}\mathcal{L}_{\text{hist}}(I,\hat{I})^2 &= SW_2(I,\hat{I})^2\\ &= \mathbb{E}_{v\sim\mathcal{U}(\mathbb{S}^{N-1})}\left[W_2(I\cdot v,\hat{I}\cdot v)^2\right]\\ &= \mathbb{E}_{v\sim\mathcal{U}(\mathbb{S}^{N-1})}\left[\left\|\text{sort}(I\cdot v)-\text{sort}(\hat{I}\cdot v)\right\|_2^2\right]\end{aligned}$$

where $\mathbb{S}^{N-1}$ is the unit sphere in dimension N, SW_2 is the 2-SWD, W_2 is the 2-Wasserstein distance, $I\cdot v$ is the scalar product between I and v, and sort is the sorting operator.

2) Band-wise Wasserstein distance $\mathcal{L}^{\lambda}_{\text{hist}}$ for a band λ

$$\begin{aligned}\mathcal{L}^{\lambda}_{\text{hist}}(I,\hat{I}) &= W_2(I_\lambda,\hat{I}_\lambda)\\ &= \left\|\text{sort}(I_\lambda)-\text{sort}(\hat{I}_\lambda)\right\|_2\end{aligned}$$

3) Gaussian Hypothesis: Anomaly detection is one of the most common analyses performed on multispectral images. A very classical multispectral anomaly detection criterion, the RX [39], makes a Gaussian assumption for the whole image, or at least for limited windows, to define a statistical test. Hence, we also measure the difference in the first- and second-order spectral statistics, using.

1) The L_2 norm between the means and covariances, respectively, $\mathcal{L}_\mu$ and $\mathcal{L}_\Sigma$

$$\begin{aligned}\mathcal{L}_\mu(I,\hat{I}) &= \left\|\mu-\hat{\mu}\right\|_2\\ \mathcal{L}_\Sigma(I,\hat{I}) &= \left\|\Sigma-\hat{\Sigma}\right\|_2.\end{aligned}$$

2) The Wasserstein distance between the two gaussian distributions given by the means and covariances which, for the sake of simplicity, we label RX even though it is not equivalent to the classical RX criterion (we do not compute a Mahalanobis distance)

$$\mathcal{L}_{\text{RX}}(I,\hat{I}) = \sqrt{(\mu-\hat{\mu})^2+\text{Tr}\left(\Sigma+\hat{\Sigma}-2\Sigma^{\frac{1}{2}}\hat{\Sigma}\Sigma^{\frac{1}{2}}\right)} \tag{11}$$

where I and $\hat{I}$ have, respectively, (μ,Σ) and $(\hat{\mu},\hat{\Sigma})$ as statistics.

4) Spectrum: Cloud field images exhibit fractal behaviors, which can be analyzed through the decay of the azimuthal modulus of the Fourier transform of the image. Which is defined as follows:

$$\mathcal{F}_{\text{rad}}(I)(r) = \frac{1}{N_r}\int_0^{2\pi}\|\mathcal{F}(I)(r\cos\theta, r\sin\theta)\|d\theta \tag{12}$$

where I is a single-channel image and $\mathcal{F}$ the Fourier transform operator. As one expects a power-law decay $\mathcal{F}_{\text{rad}}(r)\sim(1/r^p)$, the differences in fractal behaviors are quantified by the l_2 distance between the two logarithm of the azimuthal spectra

$$\mathcal{L}_{\text{sp}}(I,\hat{I}) = \left\|\log(\mathcal{F}_{\text{rad}}(I))-\log\left(\mathcal{F}_{\text{rad}}(\hat{I})\right)\right\|_2. \tag{13}$$

We compute this distance on the band-wise image I_λ for each spectral band λ and on the gray-scale image I_{mean}

$$\mathcal{L}^{\lambda}_{\text{sp}}(I,\hat{I}) = \mathcal{L}_{\text{sp}}\left(I_\lambda,\hat{I}_\lambda\right) \tag{14}$$

$$\mathcal{L}^{\text{mean}}_{\text{sp}}(I,\hat{I}) = \mathcal{L}_{\text{sp}}\left(I_{\text{mean}},\hat{I}_{\text{mean}}\right). \tag{15}$$

5) Gradients: The final features we consider are image gradients, which are key to several cloud image applications [40], [41], [42]. Gradient calculation in the two spatial directions (horizontal and vertical) on an image I, leads to two multispectral images $\nabla_x I$ and $\nabla_y I$ that can be used together to obtain the magnitude resulting in another multispectral image $|\nabla I|$. A proper reconstruction should exhibit a distribution of gradients as close as possible to the original one. Hence, we use the SWD to evaluate the deviation between the two distributions

$$\mathcal{L}_{\text{grad}}(I,\hat{I}) = SW_2\left(|\nabla I|,|\nabla\hat{I}|\right). \tag{16}$$

We further compute the Wasserstein distance over the spectral bands in a similar way

$$\mathcal{L}^{\lambda}_{\text{grad}}(I,\hat{I}) = W_2\left(\nabla I_\lambda,\nabla\hat{I}_\lambda\right). \tag{17}$$

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A total of 1000 random directions is used to compute the SWDs, $\mathcal{L}_{\text{grad}}$ and $\mathcal{L}_{\text{hist}}$.

D. Results

We show qualitative results with reconstruction examples in Fig. 4 for the S2-11B dataset and in Fig. 5 for the HyTexiLa dataset. Quantitative results are presented in Table II for the three datasets. In addition, we show the band-wise comparison for distribution, gradients, and spectral metrics in Fig. 6 for S2-11B and HyTexiLa datasets.

1) Visual Quality: Our methods enable faithful texture reconstruction for multispectral images of cloud fields in the S2-11B dataset and textures in the HyTexiLa dataset. However, the results differ depending on the method used. Indeed, we occasionally observe grid-like artifacts on the images synthesized using a projected style distance (either PCA or PCA + Color), while the ones synthesized using the stochastic style distance do not exhibit this kind of artifacts. The PCA method is the one showing the coarsest artifacts, and the use of color transfer significantly enhances the visual aspect of the synthetic images.

2) Comparison With RGB Texture Synthesis: As a sanity check, we compare our method with the result of applying the original method from [9] to the three multispectral bands that are closer to the RGB spectral content. For this, we consider bands 4, 3, and 2 for the Sentinel2 dataset and bands 5, 3, and 1 for the HyTexiLa dataset. We perform texture synthesis only using these 3 bands. Then, we compare the obtained three-channel images with the images obtained by keeping only the same 3 bands after optimizing the multispectral images using the aforementioned multispectral style distances. The quantitative comparison of the 3 RGB bands is given in Table III. The considered metrics are computed only on the RGB bands.

Without surprise, the $\mathcal{L}_{\text{style}}^{\text{RGB}}$ loss is smaller when optimizing the RGB image. Also, the stochastic style distance performs much better than the projected ones for this metric. Next, one observes that almost every proposed style distances yield a better or similar preservation of the gradient, the histogram, and the spectrum (except for *PCA* on HyTexiLa). This may be due to a better account of interband dependency when optimizing over all bands.

3) Comparison Between Methods: We now use all metrics introduced above to compare the three methods that we have introduced to perform neural multispectral texture synthesis: using the stochastic style distance, using the projected distance, and finally using the projected style distance together with the color transfer. Of course, each of these methods performs best when using the metric that has been optimized by the method, and we therefore focus on the remaining metrics to reach some conclusions. Results are gathered in Table II.

First, we observe that the stochastic style distance enforces greater respect for the Fourier spectra and the gradients in the generated textures, two features that are related to the spatial structure of the images. We can see in Fig. 6 that this is the case across all spectral bands.

For the multispectral distribution, the *PCA + Color* method stands out as it outperforms the *Stochastic* method on the SWD $\mathcal{L}_{\text{hist}}$ for Sentinel2 datasets. However, other statistics yield different conclusions. The projected style distance ensures better respect of the mean $\mathcal{L}_{\mu}$ while its stochastic counterpart seems to allow better respect of the higher order statistics, as seen for the covariance $\mathcal{L}_{\Sigma}$. Then, we also observe different dynamics across the spectral bands with the *Stochastic* method outperforming *PCA + Color* for 4 out of 11 spectral bands for S2-11B.

In addition, for S2-11B, the band-wise analysis clearly reveals that all methods underperform on the two correction bands, at 442 and 945 nm (see Fig. 6). This was expected as they present more variability than other bands. However, the two most effective methods are less sensitive to this high variability, showing a greater ability to take multispectral information into account.

When comparing the results obtained for S2-11B and S2-9B, we observe that, as the two bands that presented the more variability are removed, the synthesis task becomes easier. Indeed, those bands showed themselves to be harder to synthesize for the 11-band images and the results on the 9-band images show a quantitative improvement for both of the tested methods. The same goes when we compare the synthesis on the 3 RGB bands with the benchmark (consisting of optimizing the images only on these 3 bands) and the 11-band dataset: a fewer number of bands to consider enhance the quality of the reconstruction for both tested methods.

These results indicate the global superiority of the stochastic style distance over the projected ones both from a visual and a numerical perspective. They also highlight the effectiveness of color transfer. This supports the hypothesis put forward in Section III-D concerning the potential influence of color distribution on the quality of the descriptors provided by the network. We show that this improves synthesis quality by acting only on the colors of the images fed to the network, and not on spatial information while maintaining the same compression ratio.

E. Ablations

1) Effect of the Number of Random Triplets in the Stochastic Style Distance: In this section, we study the influence of the batch size (i.e., the number of triplets drawn at each iteration) on the synthesis results when using the stochastic style distance as a minimization objective. We perform exemplar texture synthesis on the S2-11B images and vary the batch size.

The numerical results of our experiment are given in Table IV and illustrated in Fig. 7. We also provide visual examples in Fig. 8. They show that the higher the batch size is, the better the results for all the tested metrics. We observe a great interest in increasing the batch size up to at least 5 triplets. In fact, we also observed this threshold in visual quality starting at 5 for most of the images tested. It can be seen that the synthetic image often fails to reproduce high-level textural patterns for batch sizes less than 5.

2) Effect of the Image Palette in Color Transfer: As mentioned in Section IV-B, we used three different images as a color palette in the context of the color transfer operation

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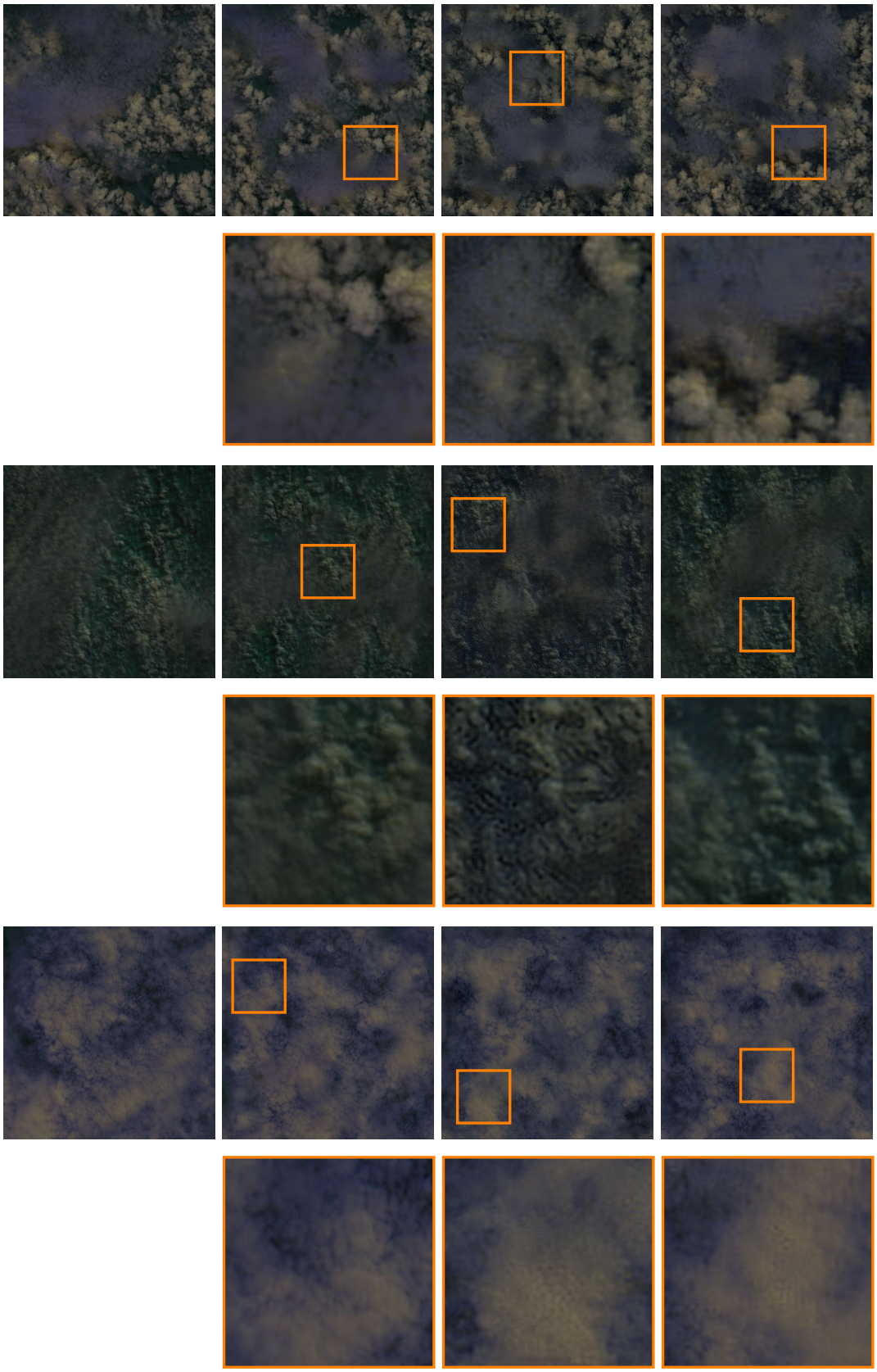


Fig. 4. Examples of multispectral texture synthesis using different style distances as the optimization objective. The images displayed are obtained by pooling the 11 spectral bands to obtain three-channel images allowing visual representation of multispectral images (channel 1: bands 1, 2, 3, 4; channel 2: bands 5, 6, 7, 8; channel 3: bands 9, 11, 12). Exemplar textures are shown on the left, and each following column corresponds to a previously presented method of synthesis. Zooms corresponding to highlighted regions are also displayed. Images that optimize the stochastic loss allow a visual appearance more faithful to the original textures than the ones minimizing a projected style distance, which may present grid-like artifacts. The introduction of color transfer after the projection strongly mitigates these defects in the generated images.

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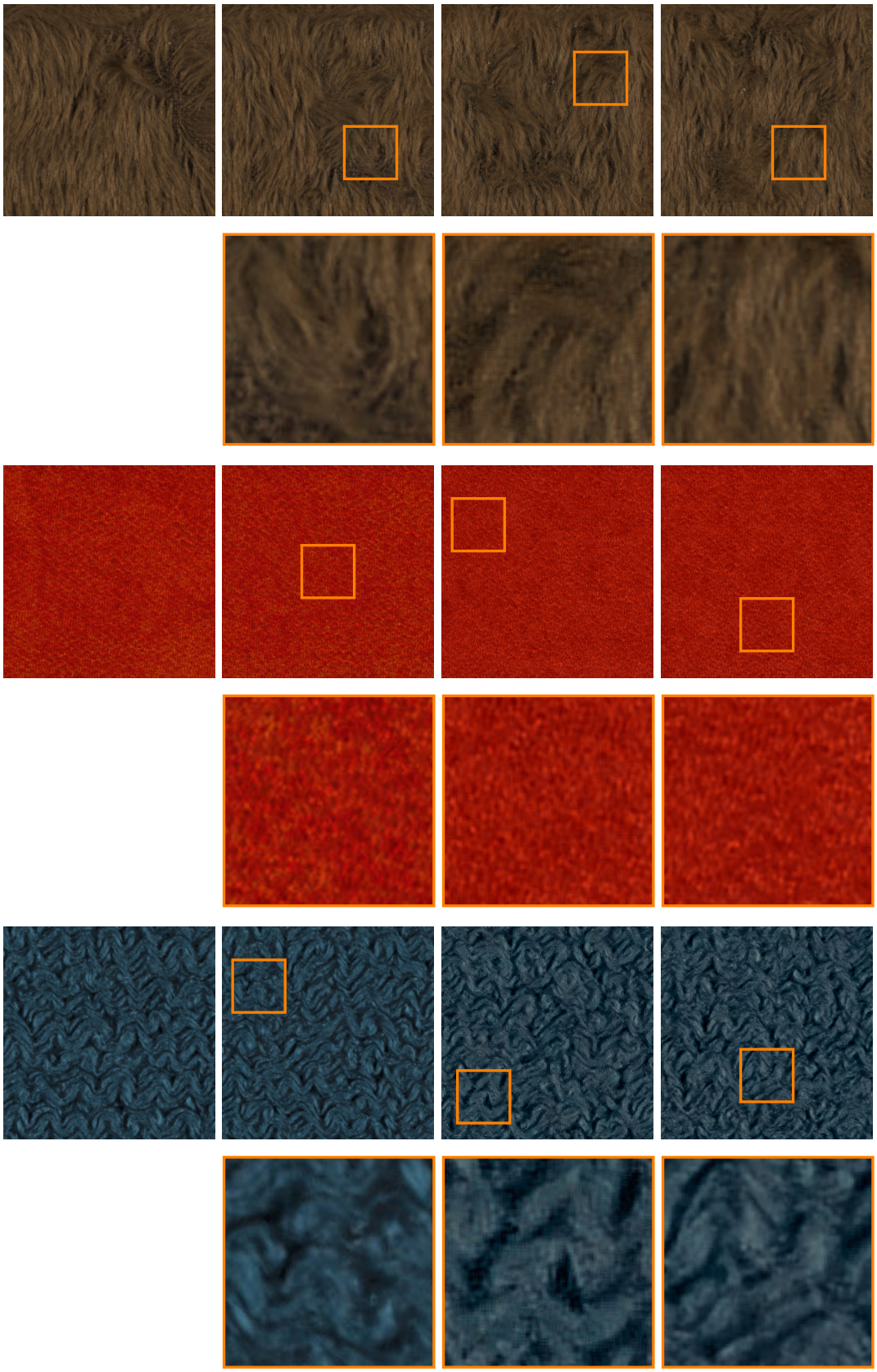


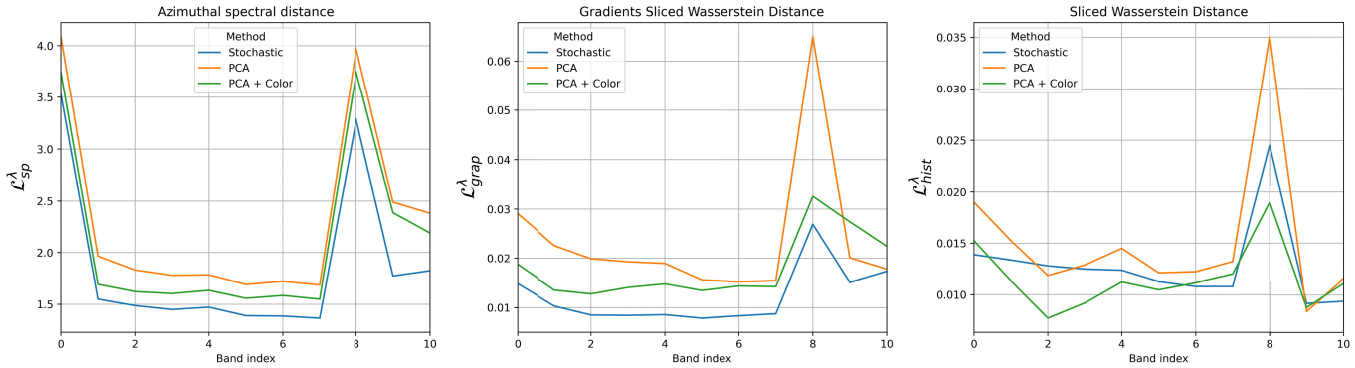
Fig. 5. Examples of multispectral texture synthesis using different style distances as the optimization objective. Exemplar textures, taken from the HyTexiLa dataset, are shown on the left, and each following column corresponds to a previously presented method of synthesis. Zooms corresponding to highlighted regions are also displayed. The images displayed are obtained by taking RGB bands. Images that optimize the stochastic loss allow a visual appearance more faithful to the original textures than the ones minimizing a projected style distance, which may present grid-like artifacts. The introduction of color transfer after the projection strongly mitigates these defects in the generated images.

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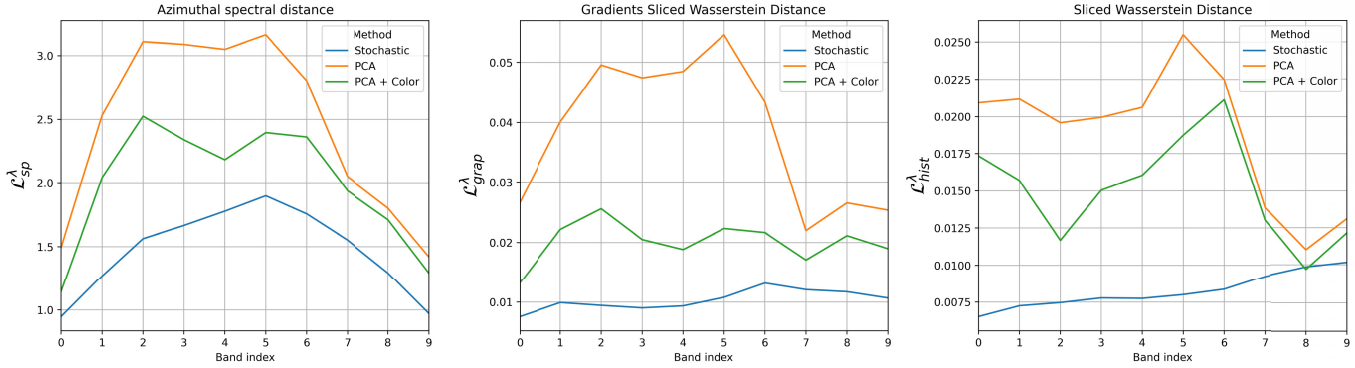
TABLE II

EVALUATION OF MULTISPECTRAL TEXTURE SYNTHESIS THROUGH CNN TRAINED ON RGB IMAGES. METRICS WERE COMPUTED AND AVERAGED OVER 130 MULTISPECTRAL IMAGES FOR S2-9B AND S2-11B DATASETS AND OVER 112 IMAGES FOR THE HYTEXILA DATASET

Dataset	Method	$\mathcal{L}_{style}^{MS} \downarrow$	$\mathcal{L}_{style}^{PCA} \downarrow$	$\mathcal{L}_{sp}^{mean} \downarrow$	$\mathcal{L}_{grad}\ (\cdot 10^{-3}) \downarrow$	$\mathcal{L}_{hist}\ (\cdot 10^{-3}) \downarrow$	$\mathcal{L}_{\mu}(\cdot 10^{-3}) \downarrow$	$\mathcal{L}_{\Sigma}(\cdot 10^{-3}) \downarrow$	$\mathcal{L}_{RX}(\cdot 10^{-3}) \downarrow$
S2-9B	Stochastic	**0.09**	161.51	**1.72**	**12.46**	9.19	14.69	**4.93**	15.27
	PCA	404.54	**0.21**	2.01	26.24	9.48	5.75	8.41	7.22
	PCA + Color	27.92	745.75	1.88	16.30	**7.66**	**3.10**	7.08	4.04
S2-11B	Stochastic	**0.17**	676.84	**1.96**	**13.52**	12.63	17.36	**14.02**	18.62
	PCA	115.38	**0.59**	2.33	26.52	15.02	12.20	21.45	15.68
	PCA + Color	68.00	4872.01	2.21	18.94	**11.70**	**4.58**	21.01	**7.87**
HyTexiLa	Stochastic	**2.55**	681.84	**1.47**	**11.61**	**9.10**	8.12	**12.59**	8.38
	PCA	3220.52	**1.57**	2.49	41.27	18.11	5.67	33.01	11.31
	PCA + Color	1174.98	2822.59	1.99	21.77	15.77	**2.33**	29.99	**7.23**



(a)



(b)

Fig. 6. Band-wise comparison of the proposed methods for Fourier spectra, gradients, and multispectral distances. (a) Sentinel2-11B. (b) HyTexiLa.

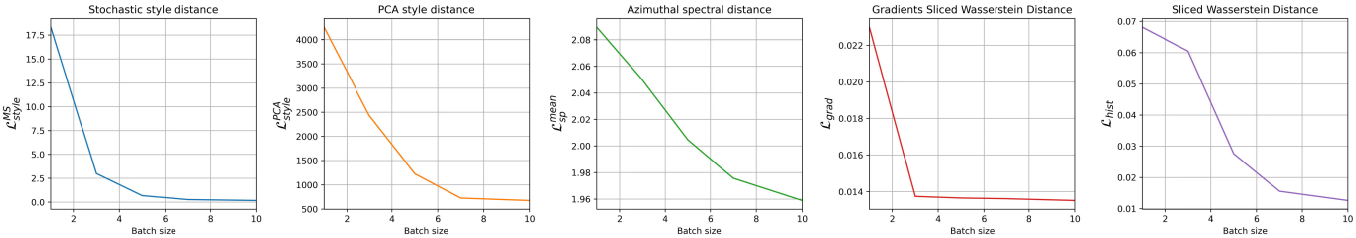


Fig. 7. Influence of the number of triplets drawn at each evaluation of the stochastic style distance on the principal multispectral texture metrics used. Metrics are averaged on 130 images.

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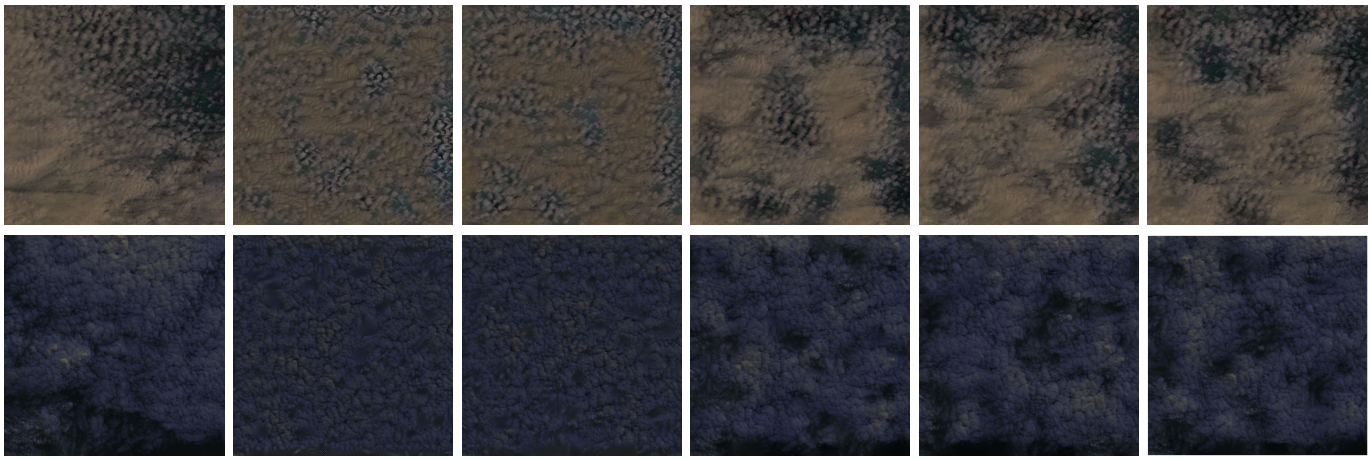


Fig. 8. Examples of multispectral texture reconstruction using stochastic style distance as minimization objective and varying the number of triplets N_B drawn at each iteration. Batch size inferior to 5 fails to reproduce high-level texture patterns in synthetic images. The images displayed are obtained by pooling the 11 spectral bands to obtain three channels (see 4 for details). Exemplar textures are shown on the left, and each following column corresponds to a batch size, labels are shown on the bottom.

TABLE III

EVALUATION OF OUR METHODS ON RGB BANDS. THE LINE RGB REFERS TO A BASELINE WHERE IMAGES ARE OBTAINED BY OPTIMIZING ONLY THE 3 RGB BANDS TO MATCH DEEP FEATURES STATISTICS. ALL METRICS ARE COMPUTED ON THE RGB BANDS. BEST RESULTS ARE HIGHLIGHTED IN BOLD, BEST RESULTS AMONG MULTISPECTRAL METHODS ARE UNDERLINED

Dataset	Method	$\mathcal{L}_{style}^{RGB} \downarrow$	$\mathcal{L}_{sp}^{mean} \downarrow$	$\mathcal{L}_{grad}$ $(\cdot 10^{-3}) \downarrow$	$\mathcal{L}_{hist}$ $(\cdot 10^{-3}) \downarrow$
S2-9B	RGB	**0.019**	2.03	28.53	14.09
	Stochastic	0.059	**1.58**	**9.32**	9.75
	PCA	565.62	1.94	20.81	9.07
	PCA + Color	31.15	1.68	11.63	**8.25**
S2-11B	RGB	**0.019**	2.03	28.53	14.09
	Stochastic	0.099	**1.68**	**9.33**	12.41
	PCA	85.15	2.09	20.55	13.02
	PCA + Color	44.71	1.87	13.46	**9.32**
HyTexiLa	RGB	**0.20**	1.96	27.80	9.82
	Stochastic	1.15	**1.66**	**9.98**	**7.87**
	PCA	2042.09	3.27	46.55	20.04
	PCA + Color	811.56	2.66	23.39	14.24

TABLE IV

MEAN PRINCIPAL METRICS RESULTS FOR MULTISPECTRAL TEXTURE SYNTHESIS ON 11-BAND SENTINEL-2 IMAGES WHEN VARYING THE NUMBER OF TRIPLETS DRAWN AT EACH ITERATION

Batch size	$\mathcal{L}_{style}^{MS} \downarrow$	$\mathcal{L}_{style}^{PCA} \downarrow$	$\mathcal{L}_{sp}^{mean} \downarrow$	$\mathcal{L}_{grad}$ $(\cdot 10^{-3}) \downarrow$	$\mathcal{L}_{hist}$ $(\cdot 10^{-3}) \downarrow$
1	18.32	4254.55	2.09	22.98	68.11
3	3.00	2435.40	2.05	13.74	60.51
5	0.68	1228.71	2.01	13.65	27.59
7	0.27	726.56	1.98	13.62	15.60
10	**0.17**	**676.84**	**1.96**	**13.52**	**12.63**

(see Fig. 9). We present and discuss the results we obtained—that are available in Table V—in this section.

We note that the results depend on the image chosen as a color palette. One of the images—*Pebbles*—performs best when used as a palette and allows for significant improvement



(a) (b) (c)

Fig. 9. Examples of natural images used as color palettes for the proposed color transfer operation. (a) Wall. (b) Pebbles. (c) Fabrics.

over the simple *PCA* style distance, for the different datasets tested. In the meantime, the *Fabrics* image shows a much more modest improvement, if any at all. These observations are consistent over the datasets, suggesting that the statistics of the palette influence the results.

V. CONCLUSION

In this article, we have introduced several strategies to generalize state-of-the-art RGB texture synthesis methods to multispectral images. The main difficulty in doing so is to generalize style distances obtained from CNN pretrained on three-channel images to a larger number of bands.

The new style distances introduced were used to perform exemplar-based texture synthesis on single multispectral images of cloud fields from the Sentinel-2 mission and on texture images from the HyTexiLa database. They show satisfactory results, both visually and numerically.

Our results and conclusions are consistent across the different datasets tested. The stochastic style distance ensures the best visual reconstruction and globally shows the best numerical results among the tested methods, but comes at a higher computational cost. The use of a projection is faster but less efficient. However, applying color transfer upstream of the projection happens to largely enhance the quality of the reconstruction.

Finally, all of the proposed methods do not require any additional training, since they leverage an RGB CNN that has proven itself to perform well for texture synthesis and

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TABLE V

EVALUATION OF MULTISPECTRAL TEXTURE SYNTHESIS THROUGH CNN TRAINED ON RGB IMAGES. METRICS WERE COMPUTED AND AVERAGED OVER 130 MULTISPECTRAL IMAGES FOR S2-9B AND S2-11B DATASETS AND OVER 112 IMAGES FOR THE HYTEXILA DATASET

Dataset	Method	Palette	$\mathcal{L}_{style}^{MS} \downarrow$	$\mathcal{L}_{style}^{PCA} \downarrow$	$\mathcal{L}_{sp}^{mean} \downarrow$	$\mathcal{L}_{grad}\ (\cdot 10^{-3}) \downarrow$	$\mathcal{L}_{hist}\ (\cdot 10^{-3}) \downarrow$	$\mathcal{L}_{\mu}(\cdot 10^{-3}) \downarrow$	$\mathcal{L}_{\Sigma}(\cdot 10^{-3}) \downarrow$
S2-9B	PCA		404.54	**0.21**	2.01	26.24	9.48	5.75	8.41
	PCA + Color	Pebbles	**27.92**	745.75	**1.88**	**16.30**	**7.66**	**3.10**	**7.08**
		Wall	276.26	427.98	2.02	18.42	10.95	7.13	8.03
		Fabrics	250.54	111.51	2.03	25.01	11.09	6.92	8.24
S2-11B	PCA		115.38	**0.59**	2.33	26.52	15.02	12.20	21.45
	PCA + Color	Pebbles	**68.00**	4872.01	**2.21**	**18.94**	**11.70**	**4.58**	**21.01**
		Wall	98.38	2158.46	2.35	21.92	14.39	9.24	21.63
		Fabrics	102.09	11008.26	2.33	29.78	15.32	12.80	22.16
HyTexiLa	PCA		3220.52	**1.57**	2.49	41.27	18.11	5.67	33.01
	PCA + Color	Pebbles	**1174.98**	2822.59	**1.99**	**21.77**	**15.77**	**2.33**	**29.99**
		Wall	1431.20	7625.62	2.02	22.12	17.61	5.96	31.69
		Fabrics	1917.30	4279.01	2.19	33.26	18.52	7.80	31.49

style transfer. The solutions we propose are not specific to texture modeling and can be applied to a wide range of style-like distances that take advantage of the capabilities of pretrained neural networks. The most direct perspective we can highlight is style transfer. A second perspective is the use of the proposed style distances for feed-forward approaches, texture synthesis, or more general restoration or editing tasks. Eventually, common distances based on neural networks such as the widely used FID [43] or LPIPS [44] can also be extended to multispectral/hyperspectral imaging with the strategies proposed in this article, opening interesting avenues for multispectral image quality evaluation. Another perspective lies in the use of the texture models presented here to train new networks to perform multitexture synthesis.

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Sélim Ollivier received the M.Eng. degree in mathematics and data science and the M.Sc. degree in fundamental and applied mathematics from CentraleSupélec, Gif-sur-Yvette, France, in 2023. He is currently pursuing the Ph.D. degree with LTCI, Télécom Paris, Palaiseau, France, and with DOTA/AILab, ONERA, Palaiseau, under the supervision of Yann Gousseau and Sidonie Lefebvre.

He is working on the development of deep learning models for texture synthesis applied to multispectral/hyperspectral and 3-D imaging.



Yann Gousseau received the B.Eng. degree from the École Centrale de Paris, Châtenay-Malabry, France, the Part III of the Mathematical Tripos degree from the University of Cambridge, Cambridge, U.K., in 1995, and the Ph.D. degree in applied mathematics from the University of Paris-Dauphine, Paris, France, in 2000.

He is currently a Professor with Télécom Paris, Palaiseau, France. His research interests include the mathematical modeling of natural images and textures, generative models, computer vision, image, and video processing.



Sidonie Lefebvre received the Ph.D. degree from Ecole Centrale Paris, Gif-sur-Yvette, France, in 2006.

She is currently a Research Scientist in statistics with Optics and Associated Techniques Department, ONERA, Palaiseau, France. Her research interests are centered on the design and modeling of computer experiments, uncertainty quantification and sensitivity analysis, and target detection.

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